SSD MEMORY DESIGN

Contents

Introduction 2

SSD/HDD Internal Category-based Memory System 3

SSD/HDD Partition Management System 4

DMA Configuration and Boot system 5

SSD/HDD Category-based Memory Security 6

Filesystem and External Drive Sorting Systems 7

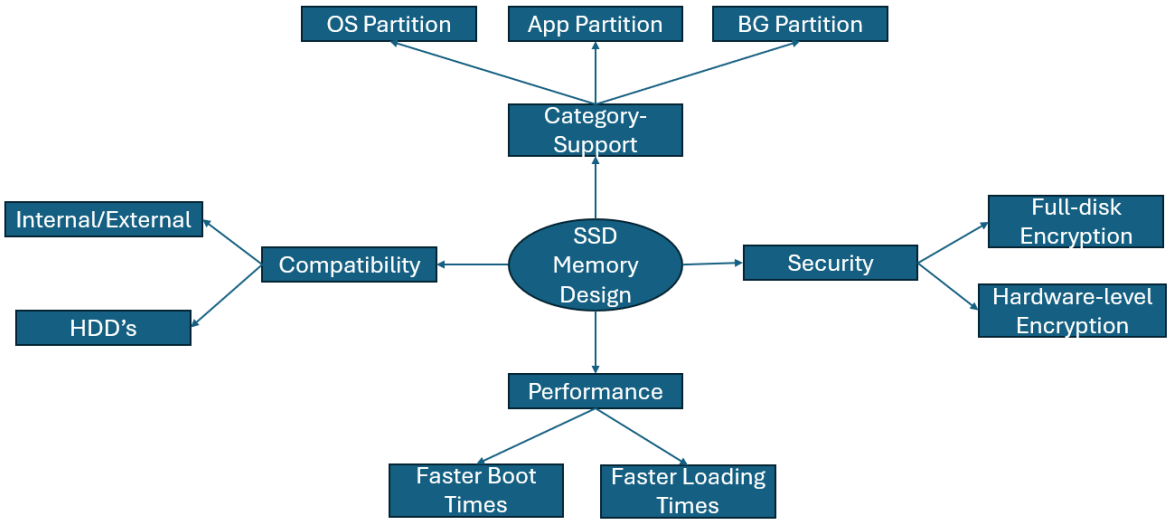
SSD Memory Design Workflow 8

SSD Memory Design Trade-Offs 9

Introduction

This SSD Memory Design is designed to improve performance on both internal and external SSDs. It is also designed to improve performance of older hardware such as internal and external HDDs. This SSD memory design was developed to allow for quicker loading times and boot times on modern computing systems that require performance and reliability. This SSD memory design does not require any direct hardware changes and can be implemented quickly through OS and minor firmware updates. This design will explain the category-based memory, the workflow and the trade-offs with migrations.

This SSD Memory Design cannot promise perfect performance or reliability. However, it can promise to improve performance through loading and boot times on modern computing systems. It can also provide compatibility with HDDs both externally and internally as this memory design is heavily managed by the operating system. It is important to remember that this memory design may be less effective on HDD's compared to SSDs. Make sure to adapt any features to the system's need.



The spider diagram above shows a visual layout of the SSD Memory Design. It shows 2-3 features of each principle involved in the design. This diagram does not show all features, and more detail can be found in the sections below.

SSD/HDD Internal Category-based Memory System

- **This category-based memory system is designed to ensure that the SSD can store its data in an organised and searchable format.**
- **This system has 4 categories in total.**
- **The first category is the operating system which will store the operating system files and the tools that are directly built into the operating system.**
- **The second category is apps which will store all user data and applications that aren't directly part of the operating system.**
- **The third category is background also known as BG which will store temporary files and other data that the OS only needs temporary storage.**
- **The fourth category is recovery which will store the recovery mode and all the recovery data that is needed to recover the device effectively.**
- **These 4 categories were chosen as they provide coverage for all kinds of data stored on a device.**
- **Each file that is categorised receives a new tag within its metadata showing what category it is in.**
- **The system uses 2 services to ensure that tagging can be done effectively in real-time and with as much accurately.**
- **The first service is the main tagging service which handles sorting the files into the correct category to ensure the correct function in real-time.**
- **This main tagging system uses both Heuristic and Behavioural systems.**
- **For heuristics, this service uses file structure, file format, owner, path and package ownership.**
- **These multiple signals ensure that the heuristic detection system is as accurate as possible as one signal is unreliable.**
- **For behavioural, the service checks what processes are using the data as each process belongs to the app, operating system or a background process.**
- **The behavioural system determines its decision by looking at which category has the most processes running on the file.**
- **If there is a conflict, then the behavioural system determines its decision by looking at what the processes are doing with the file.**
- **This conflict system has a list of what each categories processes do with files allowing to make an accurate and reliable decision.**
- **For classification, the service makes a final decision based on probability with behavioural signal providing 25% and heuristic signals being 15% each.**
- **For example, if the service thinks that the file is 75% OS while the file is 25% App then it will pick the operating system category.**
- **The second service is a validation service which ensures that the main tagging system was correct.**
- **This service checks if the category on a file is correct by checking OS error logs for their file and suggesting a classification to the main tagging system.**
- **This service is needed as the main tagging system won't always be right.**

SSD/HDD Partition Management System

- **This partition system is needed as internal drives use partitions to store different types of data together.**
- **This system will explain how categories link to the partition system that operating systems use today.**
- **There are two different types of partition systems that category-based memory can use.**
- **The first partition system is a fixed partitions system.**
- **This system creates partitions that can not be changed except by authorised operating system processes.**
- **This means that each category will be allocated a set amount of space for functioning effectively.**
- **For the OS partition, the size will be calculated by the total amount of space needed by the files as well as a 10% safety buffer.**
- **For the recovery partition, the size will be calculated by the total amount of space needed by the files as well as a 4% safety buffer.**
- **For the BG partition, the size is pre-defined at 20GB with a 5% for Windows operating systems however, this can be lowered for other operating systems.**
- **For the application partition, it gets the rest of the space after the size for the after 3 partitions have been made.**
- **It is important that this system becomes well-managed with cleaning tools to clear up unneeded data.**
- **This system is critical for applications that need predictability over adaptability and user control.**
- **The second system is a user-control partition system**
- **This system still creates the 4 main category partitions however; it also creates an additional partition.**
- **This additional partition can be used to allow the user to create their own partitions or store certain types of data within it.**
- **For OS, the size will be calculated the same way as in the fixed partition system.**
- **For recovery, the size will be calculated the same way as in the fixed partition system.**
- **For BG, the size will be calculated the same way as in the fixed partition system.**
- **For app, it will be split up to let the free space partition exist . The main app partition will have 70% of the remaining space leaving 30% for the free space.**
- **This system will not allow users to change the size of the BG, recovery and OS partitions and likely will not allow size changes for the app partition.**
- **This system is useful for general-purpose systems where users of different technical capability exist.**
- **This system can also be changed to allow users to make new partitions completely out of space in the app partition at their own risk.**

DMA Configuration and Boot system

- **This DMA configuration and Boot system is designed to ensure that the device can use the category-based structure to boot.**
- **I will be discussing the DMA 5-layer system**
- **The first layer is using a boot header to locate the bootloader within the OS partition immediately.**
- **This boot header will point to the bootloader as well as be verified with a verification hash.**
- **This verification hash will use the hash algorithm SHA-256 and be generated by the CPU or the TPM if that is available.**
- **This hash is stored on the header itself or on TPM/HSM if it is possible to do so.**
- **This layer is designed to provide fast, secure and reliable booting for the device before handing off to the operating system.**
- **The second layer is OS partition scanning.**
- **This is where the DMA will mount the OS partition and start searching for the bootloader inside of it.**
- **The DMA will mostly likely find the bootloader here as it would have been sorted into the OS category.**
- **This layer is designed to provide a backup in case of the boot header being unavailable as well as the second fastest booting method.**
- **The third layer is category structure verification.**
- **This is where the DMA checks if the category-based structure exists at all especially if the OS partition couldn't be mounted.**
- **If the category-based structure exists, the DMA will begin mounting every other partition on the drive except for the recovery partition.**
- **If the category-based structure does not exist, the DMA will begin scanning the whole drive to find the bootloader.**
- **This layer is designed to provide another backup in case the second layer also fails while also a system to use if the category-based system doesn't exist on a device yet.**
- **The fourth layer is recovery mode booting.**
- **This is where the recovery mode from the recovery partition or wherever it may be on the drive is booted and loaded.**
- **The user can use the recovery workflows and tools available to try fix the issue as much as possible.**
- **This layer is designed to provide a fourth backup in case the third layer fails which is one of the worst-case scenarios.**
- **The fifth layer is final option presentation.**
- **This is where the user is given the option to enter the BIOS, boot from a USB or shut down the device completely.**
- **This final layer is designed to ensure that there is control even the worst-case.**

SSD/HDD Category-based Memory Security

- **This SSD/HDD Category-based memory design must have layered and robust security for each layer.**
- **I will be discussing the security for 3 components involved in this design.**
- **The first component is the tagging services.**
- **It is important as if an attacker were able to exploit this, they could get write access to the OS category and control the categories causing disruption**
- **For the main tagging manager, there are 3 main security features that should be implemented.**
- **The first security feature is that other services have read-only access to the service and its activities.**
- **This is because the verification service and other OS services need to see what's its doing, but they don't need to be able to modify it.**
- **This is important because attackers will only be able to see what the service does but unable to modify it in most cases.**
- **The second security feature is signature-based verification for the tagging service to ensure that an attacker cannot use a malicious version.**
- **This will be created using Ed25519 as the signing algorithm with the operating system verifying the signature.**
- **This is important to ensure that attackers cannot use their own malicious service to sort out files allowing the operating system to work to the correct categories.**
- **The third and final security feature is isolated write-access for the categories.**
- **This means that the main tagging service will be the only service able to add and sort files into categories.**
- **This can be implemented by using the signature system and having the OS verify if the metadata has been changed by another service.**
- **The second component I will discuss is access control.**
- **For access control, there will be isolation and RBAC between categories to ensure that attackers cannot move completely through the device easily.**
- **For the OS partition, only authorised processes such as updates will have write-access to ensure that no malware can affect the operating system.**
- **For the recovery partition, the same system as used for the operating system but specific for recovery will be used.**
- **For the BG partition, it will mainly have read-only access with write access only for specific and authorised operations.**
- **This is important to ensure that malware will have more difficult in moving around the drive.**
- **The third component is data within the drive which has one main security feature.**
- **This feature is encryption which can be done using FDE, hardware-level encryption on SSD's or using individual encryption methods.**

Filesystem and External Drive Sorting Systems

- **This section will cover the systems for the filesystem which the user will interact with and the category-based sorting system for external drives.**
- **I will begin by discussing the filesystem.**
- **The file systems will this category-based memory structure will not change much visually as the changes lay logically.**
- **This is because the file systems will have to display data from different partitions depending on the data stored on these partitions.**
- **For the application partition, it can be displayed as "documents" in windows and wherever other operating systems store user data.**
- **For the OS partition, it can be displayed as "C/System drive" in windows and other operating systems that want to display the operating system files.**
- **For the BG partition, it can be displayed as "background data" in windows and other operating systems that want to give the user the ability to see this data.**
- **For the recovery partition, it will not be displayed on the operating system as users don't need to see the data within that.**
- **For custom partitions, these can be displayed alongside the system drive icon or area since it is outside of the category-based structure.**
- **This system was designed this way to provide the users with their normal experience while the category-based structure works silently in the background.**
- **This is important because users need control over their data in a simple but efficient way.**
- **I will now discuss the external drive category sorting.**
- **This system was designed as currently it only sorts internal drive however; it can also sort external drives.**
- **There are 3 partitions/categories used in this specific system.**
- **The first category is system partition which will store all the firmware and the actual configuration data for the external drive.**
- **Its size will be calculated based on the total amount of space that these files need plus a 5% safety buffer.**
- **The second category is data partition which will store all the data on the external drive that the user has transferred or downloaded on it.**
- **Its size will be calculated based on the 70% of the space remaining after the systems partition has been created.**
- **The third and final category is the free space partition which will be used if the two main partitions are full and need more space or if the user wants to create custom portions.**
- **Its size will be calculated based on how much data storage is left after both main partitions have been created.**
- **This system is presented as two options for the user allowing them to use category-based sorting or leave the drive as it is.**
- **This means the operating system needs to support both types of formats.**

SSD Memory Design Workflow

- **The workflow will show how the SSD design should work in a device and its connection to RAM, CPU, Cache and the GPU.**
- **For the first boot of the device, BIOS and other bootloader data are loaded from the hard drive without accessing any partitions as they don't exist yet**
- **The OS is then loaded from the SSD into RAM by the CPU and DMA by checking the whole disk space.**
- **After the OS is loaded, it sorts out all data on the device in the OS, App and Background partitions for quicker access.**
- **For the second boot, BIOS and other bootloader data are loaded from SSD into RAM to start up the device.**
- **This is done as the CPU and DMA can quickly access data from the OS partition leading to faster access times.**
- **OS is loaded from SSD into RAM to load the operating system into the device for the user to use.**
- **This is also done by the GPU and DMA as they can continue to quickly access data from the OS partition.**
- **This leads to faster boot times on devices especially those with NVMe SSDs for internal hard drives.**
- **The user starts a gaming application on the device after the second boot.**
- **For example, this gaming application could be GTA 6 or COD.**
- **The application is stored within the app partition where GTA-6 specific data is moved to RAM as instructions by the CPU and DMA**
- **The operating system then organises GTA-6 specific instructions into the application category to allow for the CPU to access it quicker.**
- **The most frequently used GTA-6 data and instructions are stored in L1 or L2 cache depending on the priority system that the device is configured with.**
- **The CPU can then access GTA-6 specific instructions quickly leading to faster loading times.**
- **GTA-6 3D Geometry, Images and Rendered Frames are moved to VRAM via the CPU's PCI bus.**
- **The operating system then organises the GTA-6 VRAM data into the 3 categories of Physics Model, Textures and Frame Buffers.**
- **L2 cache will prioritize Physics Models as it would be the heaviest workload of GTA 6.**
- **Each L1 cache and SMs/CUs will be dedicated to a specific category in VRAM with physics models being the main priority.**
- **Results are directly written to VRAM and then sent to the CPU for processing where they are then displayed on the user's monitor/screen.**

SSD Memory Design Trade-Offs

- **This SSD memory design has its trade-offs just like any other system I have created.**
- **There may also be more implementation challenges although they should be relatively lighter compared to the GPU, RAM and Cache designs.**
- **I have identified 5 main trade-offs for the SSD Memory Design.**
- **The first trade-off is heat.**
- **This trade-off exists since DMA and CPU can access data from SSDs more quickly which can produce more heat.**
- **This means that without heat management the SSD can overheat causing issue for the user.**
- **This trade-off can be managed by hard drive-specific cooling methods such as passive cooling and general device cooling systems.**
- **The second trade-off is first-boot setup.**
- **This trade-off exists because the organised partitions have not been created by the operating system yet.**
- **This means that boot times may be slower as the CPU and DMA can take longer to access data from the SSD.**
- **This trade-off is acceptable as it is a one-time cost and external hard drives can completely avoid this trade-off as they don't rely on boots.**
- **The third trade-off is performance inequality.**
- **This trade-off exists since SSDs especially NVMes can perform more effectively due to faster read/write speeds compared to HDDs.**
- **This means that boot times and loading times may be slower on a device with an internal HDD compared to a device with an internal SSD.**
- **This trade-off is acceptable as HDDs are accepted by users as older hardware and it is managed as the memory architecture is also available for HDDs.**
- **The fourth trade-off is OS-Reliance**
- **This trade-off exists because the OS must perform partition management, key management, and category-sorting**
- **This means that the effectiveness of this SSD memory architecture may be limited by OS limitations.**
- **This trade-off is acceptable as operating systems act as the boundary between software and hardware, and it is also managed through other systems like DMA and the CPU.**
- **The fifth trade-off Is complexity.**
- **This trade-off exists since this system requires a large number of logical changes across memory management, security and user interfaces.**
- **This means that it can be very difficult to implement leading to misconfiguration**
- **This trade-off can be managed by using phased deployment, testing and multiple teams that perform their individual assigned tasks.**